

Pluralistic Alignment via Group-Relative Policy Optimization

with a Graph-Grounded Coverage Reward

PluralTree

1 Setup and notation

Let q be an open-ended question on which people genuinely disagree. The knowledge graph $\mathcal{G}$ is a rooted tree over survey data (ROOT $\rightarrow$ topic $\rightarrow$ subtopic $\rightarrow$ question $\rightarrow$ axis $\rightarrow$ opinion), where an *axis* is a (question $\times$ demographic attribute) pair and each opinion leaf carries one subgroup’s probability distribution over that question’s answer options.

A retrieval module (the *divergence scout*) selects an anchor $a(q) \in \mathcal{G}$ and renders its subtree into a context block, giving the policy input x_q . Retrieval is executed once, at dataset-construction time: the graph encoder and the scout are **frozen** throughout this phase, so what is trained is the reasoner’s use of retrieved evidence, not the retrieval itself.

2 Graph-grounded positions

For anchor $a(q)$ we aggregate the opinion leaves of its subtree into a set of population-level *positions*

$$\mathcal{P}(q) = \{(p_1, \pi_1), \dots, (p_n, \pi_n)\}, \quad \pi_k = \frac{1}{|S_k|} \sum_{g \in S_k} \Pr_g(\text{option } k),$$

where S_k is the set of subgroups answering the underlying survey item and π_k is the mean subgroup probability of option k (its *prevalence*). Positions below a prevalence floor are discarded as noise.

This is the crux of the design: the reward is defined against *measured human survey distributions*, not against a judge model. It therefore (i) introduces no dependence on the held-out evaluation benchmark, and (ii) can be computed on any question with a graph anchor, giving effectively unlimited training signal.

3 Reward

Let y be a generated answer, segmented into position-units $u_1, \dots, u_m$ (enumerated items or sentences). Let $e(\cdot)$ be a *held-out* sentence embedder — deliberately distinct from the encoder the scout retrieves with, so the reward cannot credit merely retrieval-shaped text — and write $s_{jk} = \langle e(u_j), e(p_k) \rangle$ for unit-normalised embeddings.

Mention. Position k is *mentioned* if some unit matches it:

$$\text{MEN}_k(y) = \mathbf{1} \left[\max_j s_{jk} \geq \tau \right].$$

Depth. Each unit is attributed to its single best-matching position, $\alpha(j) = \arg \max_k s_{jk}$, and contributes its length only if that match clears τ :

$$\text{dep}_k(y) = \sum_{j: \alpha(j)=k, s_{j\alpha(j)} \geq \tau} |u_j|,$$

with $|u_j|$ the word count of unit j . Note MEN and dep use different assignment rules: a position may be *mentioned* by a unit whose best match lies elsewhere.

Coverage. A position counts only when it is both named and articulated,

$$\text{COV}_k(y) = \text{MEN}_k(y) \wedge \mathbf{1}[\text{dep}_k(y) \geq d], \quad (1)$$

for a depth threshold d . Equation (1) is the central correction relative to a mention-only reward. Empirically, an answer that names every position in a single clause each scores near zero on the target metric while a mention-only reward scores it as full recall; without the depth term the reward is blind to the dominant failure mode.

Terms. With τ the match threshold and T an effective position count,

$$R(y) = \frac{1}{n} \sum_{k=1}^n \text{COV}_k(y) \quad (\text{recall, uniform over positions}) \quad (2)$$

$$P(y) = \frac{1}{m} \sum_{j=1}^m \mathbf{1} \left[\max_k s_{jk} \geq \tau \right] \quad (\text{precision; penalises padding}) \quad (3)$$

$$V(y) = \frac{\max(0, \sum_k \text{COV}_k(y) - T)}{T}, \quad T = \exp(H(\pi)) \quad (\text{over-enumeration}) \quad (4)$$

Recall is *unweighted* even though prevalences π_k are available. The evaluation metric counts every viewpoint cluster equally; weighting recall by prevalence would train the policy to serve the majority and neglect exactly the minority positions pluralism exists to represent.

Composition. The terms combine multiplicatively,

$$r(y) = R(y) \cdot P(y)^{\lambda_p} \cdot \frac{1}{1 + \lambda_v V(y)} \in [0, 1], \quad (5)$$

rather than as subtracted penalties. Additive penalties admit a compensatory effect — the policy inflates one term to pay for another — whereas factors in $[0, 1]$ cannot be traded off, and (5) needs no clamping. We set $\lambda_v = 0$ by default: the evaluation metric is monotone in the number of clusters covered, so it cannot reward commitment, and T is derived from *graph* prevalence skew, which does not imply few *human* viewpoint clusters. The anti-enumeration role is instead served by the depth term in (1): listing many positions shallowly leaves none above d .

4 Objective

For each prompt x_q we sample a group of G completions $\{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_q)$ and score each with (5). The group supplies its own baseline, so no value network is required:

$$A_i = \frac{r(y_i) - \mu}{\sigma + \varepsilon}, \quad \mu = \frac{1}{G} \sum_i r(y_i), \quad \sigma = \text{std}(r(y_1), \dots, r(y_G)). \quad (6)$$

Writing $\rho_{i,t}(\theta) = \frac{\pi_{\theta}(y_{i,t} \mid x_q, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} \mid x_q, y_{i,<t})}$, the objective is the clipped policy-gradient surrogate with a KL anchor to the reference policy:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_i \right) \right] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}]. \quad (7)$$

A consequence of (6) worth stating: if every rollout in a group receives the same reward, $\sigma \rightarrow 0$ and the group contributes no gradient. This is benign for the per-response reward (5), whose value depends on response content, but it becomes a real hazard for any *group-dependent* reward (e.g. crediting each rollout by its marginal contribution to the group’s union coverage), where perfect complementarity and total collapse both yield symmetric rewards and hence a vanishing advantage.

5 Policy

π_{θ} is a LoRA adapter over a trainable bf16 instruction-tuned base ($\sim 7\text{B}$); the evaluation-time reasoner is a larger quantised model and is not trainable. Consequently any comparison must include a *same-base* no-training control: a policy may improve over its own base and still sit below the larger model’s untrained baseline.

6 Leakage protocol

The evaluation benchmark’s questions and its human ratings are held out absolutely. Training prompts are drawn from the graph’s own survey questions, and any question whose normalised text matches a benchmark item is dropped at dataset-construction time. The reward (5) never consults human ratings or a judge model, so no evaluation signal can enter training through the reward path.

7 Hyperparameters

Symbol	Meaning	Value
G	rollouts per prompt (group size)	8
β	KL coefficient to π_{ref}	0.04
ϵ	PPO clip range	0.2
τ	match threshold (cosine)	0.50
d	depth threshold (words/position)	60 [†]
λ_p	precision exponent	0.5
λ_v	over-enumeration scale	0
–	sampling temperature	1.0
–	LoRA rank / α	16 / 32
–	learning rate	10 ^{−6}

Table 1: [†] Provisional. d should be calibrated by regressing the probability that a viewpoint cluster is satisfied on the number of words the response devotes to it, using the benchmark’s own human-rated reference responses — a fit that requires no judge model.

Open issues

1. **Reward–metric agreement is unverified.** The advantage (6) depends only on the *within-group ordering* of rewards. If $r(\cdot)$ does not order two answers to the same question as the evaluation metric does, training optimises noise. This should be measured before training, and measured within question rather than pooled across questions.
2. **Reachability.** (7) reweights the reference policy’s existing distribution; it cannot create behaviour the base model never samples. A best-of- K sweep on the frozen base establishes whether the target behaviour is reachable at all.
3. **Construct gap.** Positions are demographic subgroup splits; the evaluation’s clusters are groupings of individuals’ open-ended perspectives. These are related but not identical, which upper-bounds how far improvements on $r(\cdot)$ can transfer.
4. **Calibration of d .** See table footnote; the reward’s absolute scale is not meaningful until d is fit.